

Grannify – Sustainable Living through Generational Knowledge

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Abstract

Household food waste remains one of the largest contributors to global waste, with private households in the EU alone generating over 50% of total food waste annually [1]. For older generations, who grew up valuing resourcefulness and sustainability, this trend is a source of deep sadness. They hold knowledge of how to preserve food, reduce waste, and live sustainably—wisdom that is increasingly lost in a world of modern convenience and disconnection between generations. This disconnect not only amplifies waste but also undermines global efforts to achieve SDG 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production). Through a qualitative user research process, two key insights emerged: younger generations often lack the time and knowledge to practice sustainable living, while older generations tend to feel increasingly lonely as they age, longing for opportunities to share their wisdom. Grannify seeks to bridge this gap by connecting generations through a dual-purpose platform, where younger individuals learn sustainable practices from older generations, while fostering community and shared purpose. The app combines gamification, practical tasks, and a community dashboard to encourage sustainable behaviors and knowledge-sharing. By preserving traditional knowledge and transforming it into actionable practices, Grannify provides a scalable solution to address waste reduction and strengthen intergenerational bonds, contributing to more sustainable, inclusive communities.

CCS Concepts

• **Social and Professional Topics;** • **Professional Topics;** • **Computing Industry;** • **Sustainability;**

Keywords

sustainable lifestyle, responsible consumption, intergenerational knowledge, intergenerational collaboration, user-centered design

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1 Introduction

Each year, more than half of the EU's total food waste comes from private households, making them a significant contributor to global waste [1]. While this statistic represents a critical environmental challenge, it also points to significant opportunities for change. The way food is valued, preserved, and consumed varies greatly across generations [2], creating both a disconnect and a potential bridge. Older generations, who lived sustainably out of necessity, view these trends with deep sadness, remembering a time when every scrap was precious. Meanwhile, younger generations, despite their enthusiasm for sustainability, often lack the knowledge or habits to effectively reduce waste [3].

This generational disconnect is not just an environmental concern but a social one as well. Through a qualitative user research process two key insights emerged. First, younger generations often lack the time and knowledge required to practice sustainable living. Second, older generations tend to feel increasingly lonely as they grow older, exacerbated by a lack of opportunities to share their wisdom and connect with others [4]. This gap contributes to food waste and risks losing 60% of skills and traditions. However, it also represents an opportunity to connect generations and transform knowledge into actionable solutions.

At the beginning of our exploration, we approached the topic of food waste with common misconceptions. Research revealed the core issue was not sharing resources but knowledge, as younger participants admitted they didn't know how to preserve or reuse food effectively. These insights shaped the foundation of Grannify, a platform designed to address the dual challenges of food waste and generational isolation. By fostering intergenerational collaboration, Grannify enables younger individuals to learn sustainable practices from older generations while giving older participants an opportunity to share their wisdom and reconnect with their communities. Through this innovative approach, Grannify not only tackles food waste but also strengthens community bonds.

This paper outlines the development of Grannify, beginning with the research that revealed the underlying challenges and opportunities. It delves into the iterative design process, key features of the platform, and the results of user testing. By bridging generations, Grannify offers an impactful solution to pressing challenges.

2 Related work

2.1 Initiatives in Europe and Germany

Efforts in Germany and across Europe have increasingly focused on reducing food waste and addressing social isolation, particularly among older adults. Private households in Germany generate over 50% of the nation's food waste. The National Strategy for Food Waste Reduction (2019) introduced initiatives like “Zu gut für die Tonne!”, offering tips on food storage, meal planning, and leftovers reuse, along with community-driven efforts such as “Deutschland rettet Lebensmittel”, which promote behavioural change and sustainable kitchen habits [5]. Loneliness among older adults is also a pressing concern, with one-third of older Germans experiencing isolation. The Strategy to Counter Loneliness (2023) includes programs like “Together for Each Other”, which organizes social activities at over 100 community centers. Multi-generational centers also foster intergenerational engagement and cultural knowledge-sharing to strengthen community bonds and reduce isolation.

2.2 Existing Solutions and Market Opportunities

Digital platforms promoting sustainability often focus on individual habits and eco-friendly practices. Ailuna¹ encourages sustainable living through small, actionable challenges, while JouleBug² uses gamified rewards to motivate eco-conscious behaviours. AWorld³ offers personalized sustainability plans and educational content for structured guidance. Platforms addressing intergenerational relationships aim to reduce isolation among older adults. Papa pairs younger “Papa Pals” with older adults for companionship and practical support and CoGenerate⁴ fosters collaboration by reimagining aging and intergenerational roles. Some initiatives focus on storytelling and cultural preservation. Generations United connects generations through shared activities, and StoryCorps preserves personal narratives to foster connections through heritage.

The initiatives and platforms explored highlight the growing recognition of food waste, loneliness, and demographic challenges in Germany and Europe. Few solutions integrate sustainability with intergenerational engagement, leaving significant gaps and opportunities for more holistic approaches that address both sustainability and social inclusion.

3 User Research

3.1 Initial Problem Framing

The project initially aimed to reduce food waste via community-based food-sharing initiatives but encountered challenges such as logistical issues and user hesitation to share with strangers. To refine the approach, 13 surveys and 18 interviews were conducted, targeting two key groups: younger university students and seniors. Among younger participants the majority, i.e 53.8% of the survey respondents admit to discarding food, with 79.8% frequently wasting fresh items like fruits and vegetables due to recurring issues like overbuying and inefficient usage. They needed practical tools for

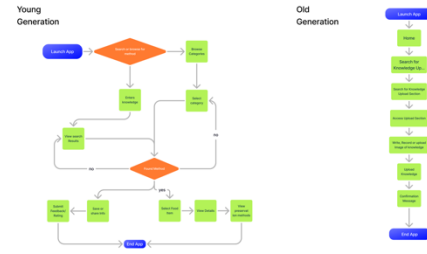


Figure 1: User flow for both target groups

waste reduction. In contrast, interviews with the elder generations gave us a key insight that 79.2% of them want to share their knowledge with the younger generations but are unable to do so. This group also highlighted loneliness and a desire for meaningful opportunities to share their experiences. Many seniors also reflected on their sustainable lifestyles in the past, noting that they rarely, if ever, used a dustbin, as every scrap of food was valued and reused. This mindset, born out of necessity, showcases their deep-rooted expertise in sustainable living practices.

3.2 Pivot to Food Preservation and Generational Collaboration

Upon identifying the core problem, we refined our approach and began exploring knowledge sharing and sustainable living as a whole. Among younger participants, 69.2% were aware of some traditional sustainable techniques but never put them into practice, while 30.8% were entirely unfamiliar with these methods—leaving none actively applying such knowledge, even when insights were shared by their elders.

The two primary barriers identified were a lack of time and insufficient knowledge. However, younger participants were motivated by the prospect of saving money and protecting the environment in the long run. This understanding became the foundation for our idea: Grannify.

4 Design Process

4.1 Findings and Design Implications

Through extensive research and interviews, we identified that the success of Grannify depends on effectively addressing the distinct needs of our two target user groups: younger users (18–45) and older adults (60+). To contrast these needs and guide our design decisions, we created two personas that embody the challenges, motivations, and goals of each group.

The personas helped us visualize the differing experiences of younger and older users through two User-Flow Diagrams, each reflecting the specific journey of each user group. These user flows were instrumental in defining the initial features of the platform, while also highlighting common elements that could benefit both groups.

¹<https://ailuna.com>

²<https://joulebug.com>

³<https://aworld.org/>

⁴<https://cogenerate.org/>

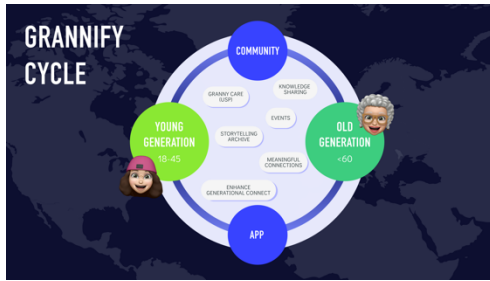


Figure 2: Grannify concept cycle representation

4.2 Ideation and Proposed Concept

After refining our initial idea, the design process for Grannify began with a collaborative ideation phase. Our team used various brainstorming techniques to generate innovative concepts within the constraints of our timeline. Mind Mapping and Crazy 8 sketches were pivotal in this process, enabling us to quickly explore and organize diverse ideas. These methods helped us visually represent complex concepts and pinpoint the most promising features for the platform.

Mind Mapping allowed us to connect different ideas and features, forming the foundation for the Grannify Cycle—a dynamic ecosystem that links younger and older users through shared knowledge and sustainable practices.

From this ideation phase, the Grannify project was born. Recognizing the distinct needs of both user groups, we designed two tailored versions of the app: one optimized for younger users seeking fun and engaging ways to adopt sustainable habits, and another focused on older users, providing an intuitive platform to share their skills and experiences while staying socially connected. This dual approach enhances accessibility and bridges technological barriers, ensuring inclusivity for all users.

Grannify’s value proposition lies in its ability to transform the timeless wisdom of older generations into actionable insights for younger users. By integrating Human-Computer Interaction (HCI) principles, the platform prioritizes usability, accessibility, and engagement, creating a seamless user experience for both generations. Through this human-centered design approach, Grannify is positioned as a powerful tool for fostering sustainable communities and meaningful connections on both local and global scales.

4.3 Prototyping and Testing

4.3.1 First Phase: Paper Prototyping. The first iteration of Grannify employed low-fidelity paper prototypes to visualize early concepts and evaluate core platform functionalities, with a focus on two key user groups: younger users and older adults.

For younger users, the design incorporated gamification to make sustainability both fun and educational. Features like task streaks, progress tracking, and the Granny Care module, where users could complete tasks such as preparing herbal tea or learning traditional techniques, were included to promote engagement. The aim was to blend entertainment with sustainability, encouraging intergenerational connections.

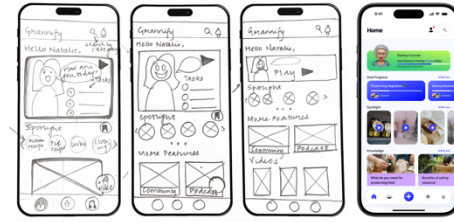


Figure 3: Paper prototype iterations and final design for home screen

For older adults, the design emphasized simplicity, accessibility, and empowerment. The platform allowed users to share knowledge through text, voice recordings, and images, enabling them to document and share traditional practices. Opportunities for community engagement were also provided, alongside a tablet-friendly interface to ensure ease of use. The goal was to create an inclusive space for older users to share their expertise while fostering connections with others.

The clickable mobile paper prototype was tested with 6 participants aged 20–28, who interacted with the platform and provided feedback in real time. Participants enjoyed the gamified elements and sustainability tasks but suggested improving visual clarity in progress tracking and making content more concise to align with their fast-paced lifestyles.

A tablet version was tested with 4 participants aged 68–90, who received guidance throughout the session. These participants appreciated the simplicity of the interface and the tools for knowledge sharing. However, they suggested incorporating more real-world engagement opportunities, such as community events, to foster in-person connections.

Overall, the testing revealed two key insights. First, integrating digital features with real-world interactions is crucial, especially for older users who value face-to-face engagement. Second, ensuring accessibility and usability across age groups is essential for promoting inclusivity and confidence in all users.

4.3.2 Second Phase: High-Fidelity Prototype. The second iteration focused on translating these insights into a high-fidelity prototype. Created in Figma, this version was tested with 17 participants, including 10 younger users (ages 18–34) and 7 older users (ages 60–90). For older users, we refined the navigation by enlarging fonts and using more recognizable icons, ensuring a seamless user experience. Step-by-step onboarding, and guidance through the knowledge sharing tasks. Tooltips were introduced to enhance intuitiveness. Additionally, features like event descriptions and RSVP options encouraged offline participation and community engagement. For younger users, we expanded the gamification elements, incorporating progress bars, milestone tracking, badges, and social components to keep them engaged. The Granny Care tasks were broadened to include a wider variety of activities, ensuring that the platform remained engaging over time. Social features were also added, allowing users to share achievements and compare progress with friends.

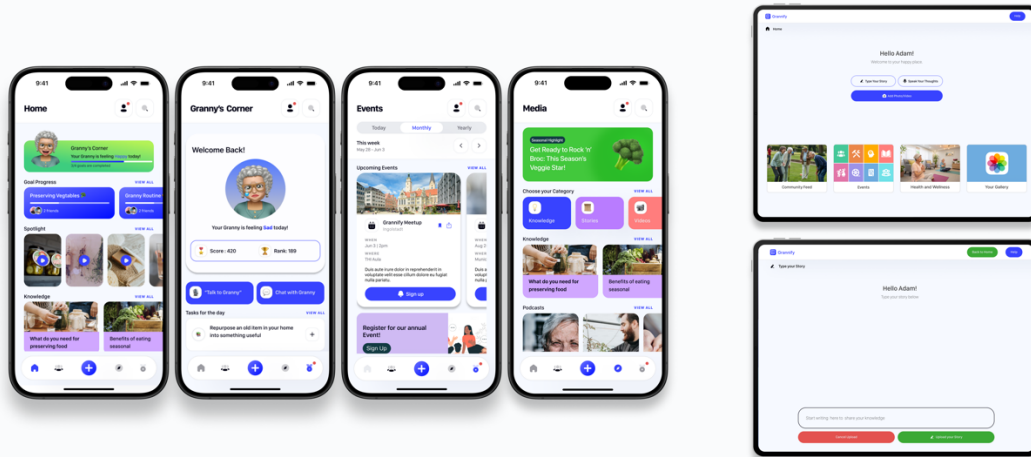


Figure 4: Final prototype for both mobile and tablet version of the application

The high-fidelity prototype testing provided further insights. Younger users appreciated the gamification but suggested that video content be shortened to better fit their fast-paced routines. Older users found the updated navigation and onboarding helpful, but many felt they needed more guidance on how to share their knowledge, particularly when it came to recording and uploading content. In response, we simplified the onboarding process for older users, incorporating guided tutorials and clearer visuals. Additionally, we expanded the Granny Care tasks for younger users, linking their progress to interactions with older participants. To enhance community engagement, we began exploring potential partnerships with local organizations to offer real-world events and activities.

These refinements helped shape a more user-centered design, addressing the specific needs of both younger and older users while fostering an engaging and collaborative experience.

5 Final design

Grannify is a thoughtfully designed platform that seamlessly integrates the needs of both younger and older generations, fostering sustainability and promoting intergenerational collaboration. The app offers two tailored versions—one optimized for mobile devices for younger users and another for tablets to accommodate the older demographic—ensuring an intuitive, engaging experience for all users.

The design is grounded in key Human-Computer Interaction (HCI) principles, particularly usability, accessibility, and engagement. For usability, the interface is streamlined and easy to navigate, ensuring that both user groups can interact with the platform effortlessly.

5.1 App Design for Younger Users

Grannify's mobile version is designed for younger users, aged 18–45s, offering an intuitive, mobile-first interface that prioritizes accessibility and quick interaction with the platform's features. This version integrates engaging elements such as Granny's Corner, a

virtual assistant that tracks sustainable actions, provides encouragement, and offers playful feedback to keep users motivated. The app includes fun, goal-oriented challenges, such as reducing food waste or creating home gardens, encouraging users to adopt sustainable habits in an interactive and enjoyable way.

Younger users also have access to a knowledge Hub that houses sustainability techniques contributed by older users, including practical advice on food preservation, gardening, and DIY repairs. Skill Requests allow younger users to directly ask for advice from contributors based on their specific needs. In addition, the Community Section offers interactive spaces where users can join interest-based groups—such as gardening, cooking, or wellness—and engage in discussions, skill sharing, and social activities. Features like progress tracking, badges, and milestone achievements further encourage engagement by recognizing users' progress and efforts toward sustainable living.

5.2 App Design for Older Users

Grannify's tablet version is specifically designed for older users, aged 60 and above, with a focus on simplicity, accessibility, and ease of interaction. The interface is clean, with larger fonts and clear navigation to accommodate users who may have limited technical experience. Knowledge Sharing allows older users to document and share their wisdom on sustainable living through various formats such as video, audio, and text. This feature empowers older users to contribute their knowledge to the platform in a way that suits their preferences and comfort levels.

The Community Section in the tablet version enables older users to participate in social activities, engage in skill-sharing groups, and connect with others around shared hobbies and interests, such as gardening, cooking, and wellness. To further support content creation, Guided Tutorials offer step-by-step instructions to help older users upload and share their knowledge effectively, ensuring that they can contribute with minimal technical barriers. This version is designed to make it easy for older users to remain socially connected, share their experiences, and stay engaged in a supportive, user-friendly environment.

6 Future Scope

Future development of Grannify will focus on real-world evaluation, adoption strategies, and AI-driven innovations. Pilot studies in different communities will help assess its effectiveness in reducing food waste and fostering intergenerational connections. Key metrics, such as user engagement and behavioral changes, will guide improvements.

To enhance adoption, strategic partnerships with local organizations and universities will be explored, along with refinements in accessibility and gamification. Addressing digital literacy challenges for older users will be a priority. AI integration can further enrich the platform by generating content from historical books and films, offering personalized recommendations, and enabling voice-assisted knowledge sharing.

A structured roadmap will ensure continuous progress. Grannify 1.0 will focus on core functionalities, while version 1.5 will expand real-world applications. Grannify 2.0 will incorporate AI-driven features and scale globally, making the platform a long-term solution for sustainable knowledge-sharing and community building.

7 Conclusion

Grannify unites younger and older generations to tackle food waste and social isolation, empowering users through gamified learning and meaningful connection. From October to mid-January, we refined the app through an iterative design process, ensuring accessibility, inclusivity, and a seamless user experience. Beyond the app, Grannify bridges digital engagement with real-world impact, fostering eco-conscious habits and intergenerational understanding. This

journey has been deeply personal, showing us how collaboration can inspire real change. Grannify is more than an app—it's a first step toward a sustainable and connected future we're proud to help create.

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